AI Changes the Delivery System and the Product Itself

New capabilities change what we can delegate.





HOW WE BUILD

Judgment during delivery

WHAT WE BUILD

Judgment at runtime

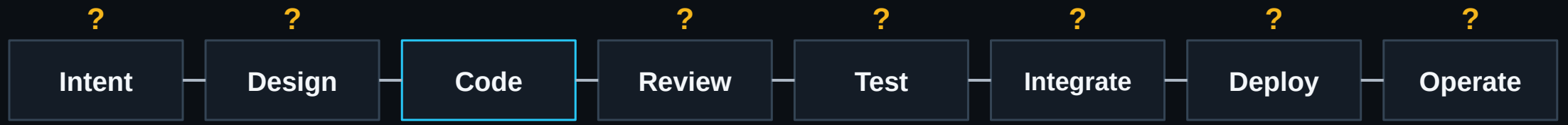
Vitalii Oborskyi Head of Delivery & PMO · Author of Uncertainty Architecture

DevCon Romania · 4 November 2026

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We Accelerated a Step. Not the System.

AI can affect every stage. Downstream gains are not automatic.



Illustrative team: changes per week; comparable work, fixed downstream capacity.

	Code	Review	Integrate	BACKLOG
BEFORE Balanced flow	8	8	8	0 no accumulation
AFTER AI Other stages have not sped up	100	8	8	+92 unfinished / week

THEORY OF CONSTRAINTS

Push system: no WIP limit, no discarded work.

Delivery stays at 8/week. WIP grows; overload can reduce quality.

The Evidence Is Already Messy

Different studies measure different outcomes — not one universal productivity effect.

NBER · SEP 2026 · >500K GITHUB DEVS

25.5× changed LOC

3.4× commits

1.3× releases

Cumulative across three tool generations

Changed LOC = additions + deletions.
Fig. 1 rounded; async releases not estimated.

More apps. Low-usage shares rise.

iOS: <10 ratings **≈78% → 87%**

Chrome: <10 downloads **19% → 33%**

4 marketplaces; iOS / Chrome shown.
Jan 2025–Apr 2026; first 3 months.

DORA · 2025 · n = 4,867

Respondents reporting improvement:
Productivity >80% · Code quality 59%

Higher adoption ↔ throughput + instability

Instability: failed production changes + unplanned fixes.

Survey association; no causal estimate.

GITCLEAR · SHARES OF CHANGED LINES

Metric	2021[1]	2026[2]
Moved code	24.65%	3.8% YTD
Copy/paste	8.66%	15.7% H1

[1]2025 report [2]2026 report; different samples.

Snapshots only; no comparable trend or causal estimate.

METR · MAY 2026 · n = 349

1.4–2× self-reported work value

Medians across three questions.

Technical workers; no causal estimate.

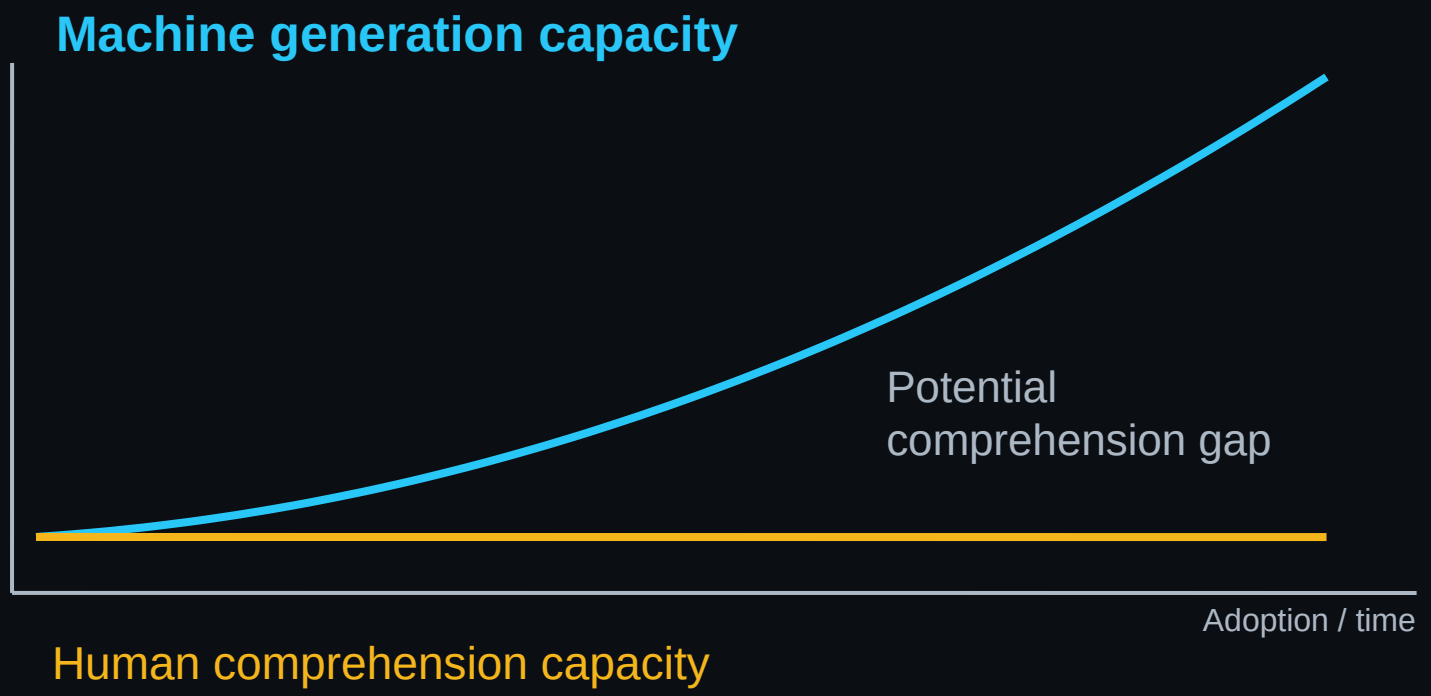
AGARWAL · JAN 2026 · 1,197 REPOS

Cognitive complexity after agent adoption

Agent-first +34.85% · Prior AI-IDE +42.87%

Matched controls; 19,236 repo-months.

The New Scarcity Is Human Comprehension



Can we explain the change?

Can we maintain the dependency?

Can we recover the system?

Illustrative overload hypothesis; motivated by the evidence, not measured by those studies.

If AI cannot recover the system, the team still must.

We Are Searching for a New SDLC Equilibrium

CAPABILITIES CHANGE

Generation
Review
Integration
Understanding

→

OPERATING MODEL MUST ADAPT

Work allocation
Ownership
Learning
Feedback

→

WORKING HYPOTHESES

Which constraint does this address?
What must be true for it to work?
What would make us revise it?

Test delivery, quality, full cost and team health.

Proposed operating models remain hypotheses about this context.

When Software Starts Thinking

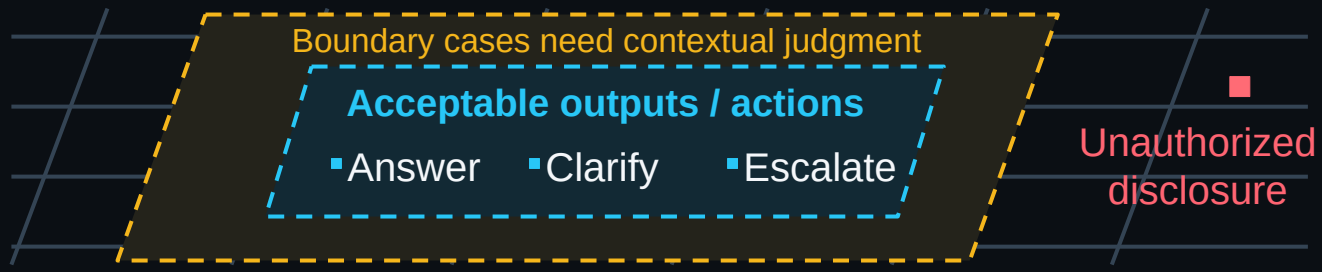
A system where consequential runtime responsibility partly depends on probabilistic Model Judgment.

IT ASSISTANT

“Help employees recover access.”

Approved guidance and permitted actions

Intents · Context · Data · Permissions



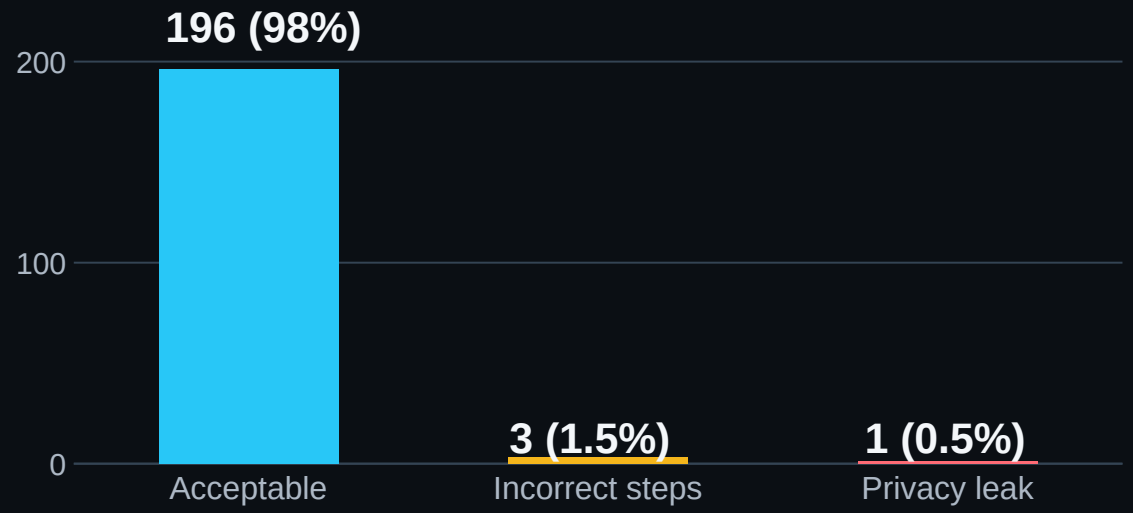
Prohibited: unauthorized disclosure

Schematic behavioral space; limits are product-specific.

Requirements define acceptable behavior and prohibited effects.

Evidence Changes the Release Decision

ILLUSTRATIVE INDEPENDENT SAMPLE · n = 200



196 acceptable responses

3 incorrect instructions
Rework

1 privacy leak
Critical breach

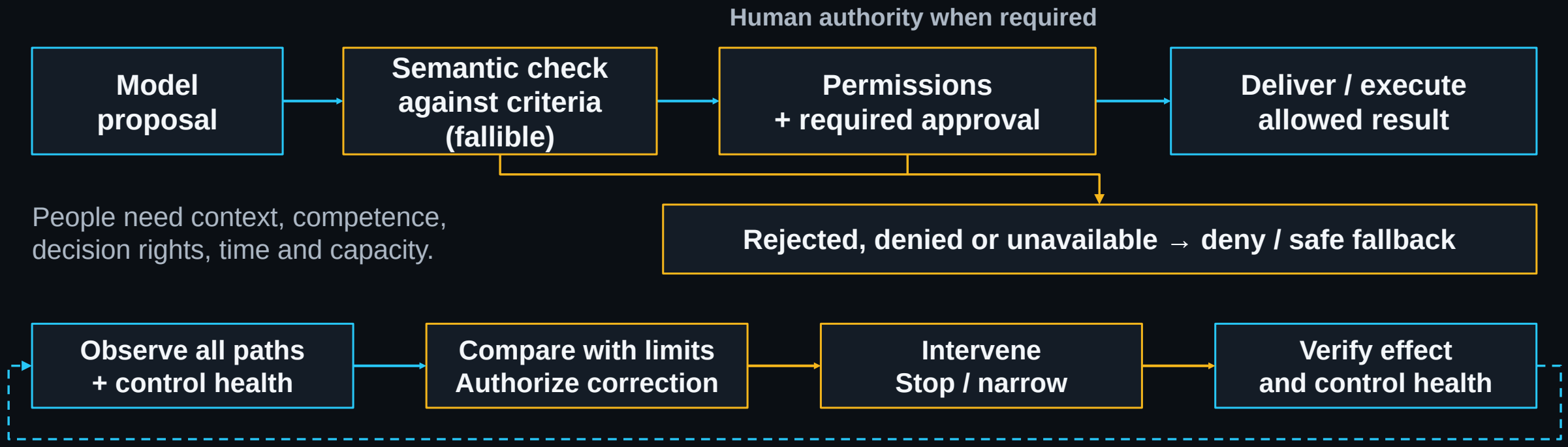
BLOCK RELEASE

A low average error rate does not excuse a prohibited disclosure.

Readiness includes behavior, evidence and an operable control path.

Control Is a Sociotechnical Capability

Evidence against agreed limits | **Authority to decide** | **Capability to act**



People need context, competence, decision rights, time and capacity.

Validate control effectiveness. Telemetry alone does not establish correctness.

Check access before disclosure. Feedback cannot undo an irreversible effect.

The Roles Move With the System

AI redistributes cognitive work across the sociotechnical system.

HOW WE BUILD	WHAT WE BUILD
Review + integration capacity	Evaluation + response capacity
Understanding + recovery	Behavioral criteria + authority
Delivery feedback	Outcome + incident feedback

SHARED DEPENDENCIES

Knowledge · Decision rights · Time and capacity · Feedback

Required decisions, evidence and capabilities belong in the delivery dependency plan.

Engineering Rigor Moves — It Doesn’t Disappear

THE LABORATORY STAYS OPEN



Which imbalance does this address?

By what mechanism?

What would show system-level improvement?

What would make us revise the hypothesis or delegation boundary?

More capability still requires engineering responsibility.

Continue the Conversation

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Research · Articles · Talks

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